

Seunghyeon Lee (이승현 · Clay Lee)

Ph.D. Candidate (expected Aug 2026) · Forest Structure & Remote Sensing (LiDAR / GEDI)

Landscape & Ecological Planning Lab, Seoul National University, Seoul, Republic of Korea

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RESEARCH INTERESTS

Airborne & spaceborne LiDAR (GEDI) for vertical forest structure · forest typology and stratification metrics (FHD, PAI, layer indices) · biomass and carbon estimation · urban ecosystems and green infrastructure · UAV/thermal wildlife monitoring · reproducible, code-driven remote-sensing science.

EDUCATION

Ph.D. Candidate, Interdisciplinary Program in Landscape Architecture

2022.03 — Aug 2026 (expected)

Integrated Major in SmartCity Global Convergence · Seoul National University · Landscape & Ecological Planning Lab · Advisor: Prof. Youngkeun Song

- Dissertation: “LiDAR-Based Characterization of Vertical Structure and Disturbance in Temperate Forests”
- Committee: Dongkun Lee, Youngryeul Ryu (Seoul National Univ.); Hansoo Kim (Gyeonggi Research Institute); Dennis Heejoon Choi (Dankook Univ.); Youngkeun Song (advisor, SNU)

M.A., Landscape Architecture

2020.03 — 2022.02

Integrated Major in SmartCity Global Convergence · Graduate School of Environmental Studies, Seoul National University · Advisor: Prof. Youngkeun Song

- Thesis: “Feasibility of Wildlife Detection Using UAV-derived Thermal and True-color Imagery”

B.S., Landscape Architecture

2011.03 — 2017.02

College of Agriculture & Life Sciences, Kyungpook National University, Daegu, Republic of Korea

RESEARCH EXPERIENCE

Visiting Research Intern — Purdue University, FNR FUSE Lab (PI: Prof. Brady Hardiman)

2023.12 — 2024.02

West Lafayette, IN, United States · Forest structure & LiDAR

Visiting Research Intern — Virginia Tech (PI: Prof. Jaeyoung Ha)

2024.07 — 2024.08

Blacksburg, VA, United States · Landscape architecture

Graduate Researcher — Landscape & Ecological Planning Lab, Seoul National University

2020.03 — Present

National R&D and municipal projects on LiDAR forest structure, carbon, and urban ecology

Research Assistant — Architecture & Urban Research Institute (AURI)

2019.08 — 2019.11

Republic of Korea

PEER-REVIEWED PUBLICATIONS

APA 7th. Name in bold = author; † = first author. Impact Factor / quartile shown where available. (SCI then KCI; newest first)

1. Kim, J., Han, S., Yun, J., **Lee, S.**, & Song, Y. (2026). Ecological structures and terrestrial insect diversity across successional stages in abandoned paddy fields. *Agriculture, Ecosystems & Environment*, 399, 110172. <https://doi.org/10.1016/j.agee.2025.110172> · IF 6.4, Q1 (2026)
2. **Lee, S.†**, Ha, U., Lee, H., Choe, H., Kim, H., & Song, Y. (2026). Multi-scale forest typologies using novel vertical layer metrics from airborne LiDAR in temperate mixed forests. *Ecological Indicators*, 185, 114798. <https://doi.org/10.1016/j.ecolind.2026.114798> · IF 7.4, Q1 (2026)
3. Shao, J., Choi, D. H., Liu, J., Tian, X., Thapa, B., **Lee, S.**, Habib, A., & Fei, S. (2026). A three-stage framework for stand-level automated stem volume estimation in temperate forests using Mobile Laser scanning. *Remote Sensing of Environment*, 335, 115246. <https://doi.org/10.1016/j.rse.2026.115246> · IF 11.4, Q1 (2026)
4. Yun, J., **Lee, S.**, & Song, Y. (2025). Assessing *Corvus frugilegus* (Rook) habitat preferences through flock-size-specific species distribution modeling using citizen science data. *Global Ecology and Conservation*, 63, e03866. <https://doi.org/10.1016/j.gecco.2025.e03866> · IF 3.4, Q1 (2025)
5. **Lee, S.†**, Song, Y., & Kil, S.-H. (2021). Feasibility Analyses of Real-Time Detection of Wildlife Using UAV-Derived Thermal and RGB Images. *Remote Sensing*, 13(11), 2169. <https://doi.org/10.3390/rs13112169> · IF 5.0, Q1 (2022)
6. **Lee, S.†**, & Song, Y. (2026). Forest-Type and Seasonal Analysis of GEDI-ALS Canopy Height Agreement and GEDI Structural Metrics (FHD, PAI): A Case Study of Forests in Gwacheon-si and Uiwang-si. *Journal of the Korean Society of Environmental Restoration Technology*, 29(2), 21–40. <https://doi.org/10.13087/kosert.2026.29.2.21>

7. **Lee, S.†**, Jang, G., Song, Y., & Kil, S.-H. (2026). Comparing LST Estimation Methods under Matched Spatiotemporal Conditions in Low- and High-Rise Residential Areas: Satellite, Simulation, and UAV Approaches. *Journal of the Korean Society of Environmental Restoration Technology*, 29(2), 41–59. <https://doi.org/10.13087/kosert.2026.29.2.41>
8. Jekal, G., Yang, Y., **Lee, S.**, & Song, Y. (2025). Non-destructive Carbon Storage Estimation of Salix Spp community In Wetlands Using LiDAR: A Case Study of Ungok Wetland. *Journal of the Korean Society of Environmental Restoration Technology*, 28(6), 95–105. <https://doi.org/10.13087/kosert.2025.28.6.95>
9. Han, Y.-H., Shin, W.-H., Kim, J.-H., Kim, D.-H., Yun, J.-W., Yi, S.-Y., Kim, Y.-H., **Lee, S.**, & Song, Y. (2024). Diel Activity Patterns of Water Deer (*Hydropotes inermis*) and Wild Boar (*Sus scrofa*) in a Suburban Area Monitored by Long-term Camera-Trapping. *Journal of the Korean Society of Environmental Restoration Technology*, 27(2), 55–65. <https://doi.org/10.13087/kosert.2024.27.2.55>

MANUSCRIPTS UNDER REVIEW

APA 7th. Name in bold = author; † = first author. (5 manuscripts; first-author listed first)

1. **Lee, S.†**, Choi, D. H., Song, Y., & Thorne, J. H. (2026). *Spaceborne LiDAR reveals post-fire vertical structural loss in a dense temperate Asian conifer plantation* [Manuscript under review]. *Science of Remote Sensing*. · 2026.04
2. **Lee, S.†**, Kim, H., & Song, Y. (2026). *Systematic bias in urban-forest vegetation coverage assessment: the influence of topo-edaphic conditions on discrepancies between ALS and field surveys* [Revise and resubmit, 1st round]. *Geoscience Letters*. · 2026.06
3. **Lee, S.†**, Kim, Y., Kim, D., Kim, H., & Song, Y. (2026). *Bi-temporal ALS assessment of vertical–horizontal forest structures and their structural association in a temperate urban forest* [Revise and resubmit, 2nd round]. *Forest Science and Technology*. · 2026.05
4. Kim, Y., **Lee, S.**, Shin, W., & Song, Y. (2026). *Integrating UAV-derived habitat metrics with movement persistence and species distribution models to characterize seasonal habitat use of invasive turtles in urban wetlands* [Revise and resubmit, 1st round]. *Global Ecology and Conservation*. · 2026.05
5. Jekal, G., Kim, Y. H., **Lee, S.**, Yun, J. W., Kim, D. Y., & Song, Y. (2026). *Monitoring transition-zone dynamics of a *Phragmites communis*–*Suaeda japonica* mosaic from multi-season Sentinel-2 in a coastal wetland* [Revise and resubmit, 1st round]. *Journal of Coastal Conservation*. · 2026.02

BOOKS

1. YOZMDOSI collective (**Seunghyeon Lee**, contributing author) (2021). *New Normal City* (뉴노멀시티). Urban Issue. ISBN 9791197343100.

INVITED TALKS

- **Seunghyeon Lee** (2025). The understanding and applications of thermal sensors. *Guest lecture, SmartCity Industrial Technology Seminar* (3 cr.), Seoul National University.
- **Seunghyeon Lee** (2025). Understandings patial information and research cases. *Guest lecture, Environmental Conservation & Land Management* (3 cr.), Seoul National University.
- **Seunghyeon Lee** (2024). Remotely sensed smart-city ecological value maps. *NEF (Neocity Empowerment Forum)*, Orlando, FL, US.
- **Seunghyeon Lee** (2024). Invasive species monitoring development and its application. *Asia Week, Fukuoka, Japan*.

CONFERENCE PRESENTATIONS (INTERNATIONAL)

First author / presenter unless marked “team”. 10 domestic presentations (KOSERT, KSEE, KILA) available on request.

- **Seunghyeon Lee**, Hansoo Kim, Youngkeun Song (2025). Novel LiDAR indices reveal scale-dependent vertical structure and typologies in a temperate forest. [Poster] *AGU Fall Meeting, New Orleans, US*.
- **Seunghyeon Lee**, Hansoo Kim, Youngkeun Song (2024). Quantifying the number of forest stratification layers of city-scale forest and characterizing by forest types. [Poster] *AGU Fall Meeting, Washington D.C., US*.
- **Seunghyeon Lee**, Hansoo Kim, Youngkeun Song (2024). Canopy layers distribution by forest patch and species at the city-level forest using airborne LiDAR. [Oral] *ICLEE, Kitakyushu, Japan*.
- **Seunghyeon Lee**, Hansoo Kim, Youngkeun Song (2024). Characterizing forest vegetation stratification by forest biotope types using airborne LiDAR. [Poster] *ESA Annual Meeting, Long Beach, CA, US*.
- BK21 SmartCity Team (2024). Smart city & green infrastructure. [Video, team] *CES (Consumer Electronics Show), Las Vegas, NV, US*.
- Landscape & Ecological Planning Lab (2023). Property-information analysis of Gyeonggi-do Province biotopes. [Oral, team] *JCK Symposium (18th), Kyoto, Japan*.
- **Seunghyeon Lee**, Youngkeun Song (2023). Wildfire-driven forest vegetation height change comparison. [Poster] *JCK Symposium (18th), Kyoto, Japan*.
- **Seunghyeon Lee**, Youngkeun Song (2023). The impacts of fire-induced disturbances on tree height and structure using GEDI-derived variables. [Poster] *ESA Annual Meeting, Portland, OR, US*.

- **Seunghyeon Lee**, Youngkeun Song (2022). Monitoring structural change of fire-induced forest vegetation using GEDI. [Oral] ICLEE (12th), Online.
- **Seunghyeon Lee**, Youngkeun Song (2022). Comparison of GEDI and aerial laser scanning datasets according to leaf-on and leaf-off seasons. [Oral] AGU Fall Meeting, Chicago, IL, US / Online.
- **Seunghyeon Lee**, Dennis Heejoon Choi, Youngkeun Song (2022). Classification of tree species and forest floor using an XGBoost algorithm with forest maps, airborne LiDAR, and satellite imagery. [Video] World Forestry Congress (15th), Seoul, Republic of Korea.
- **Seunghyeon Lee**, Sung-Ho Kil, Youngkeun Song (2022). Feasibility analyses of real-time detection of wildlife using UAV-derived thermal and RGB images. [Poster] World Forestry Congress (15th), Seoul, Republic of Korea.
- **Seunghyeon Lee**, DaeYeol Kim, Dennis Heejoon Choi, Hansoo Kim, Youngkeun Song (2021). Detecting individual broad-leaved trees by a trunk-extraction method using leaf-off airborne LiDAR. [Poster] AGU Fall Meeting, New Orleans, US / Online.

PATENTS (REPUBLIC OF KOREA)

7 registered, 4 pending.

Registered

- System and method for providing a tree-management platform. Registered 2025.09 · Sole inventor
- Method and system for determining wetland vegetation structure using drone LiDAR. Registered 2025.04 · Seoul National University
- Image-based roadside-tree information management system and method. Registered 2024.11 · Sole inventor
- Prediction of missing location points of a wildlife GPS tracker by combining in-situ and biological information. Registered 2023.07 · Seoul National University
- Self-heating target module for thermal-camera performance testing and precision data acquisition. Registered 2022.10 · Seoul National University
- Automated wildlife detection method and device using drone-mounted thermal-camera imagery. Registered 2022.05 · Seoul National University
- A method to derive an optimal plan for outdoor thermal-comfort mitigation. Registered 2022.04 · Seoul National University

Pending

- Quantitative assessment and automatic classification of wildfire damage using spaceborne LiDAR. Pending 2025 · Dankook Univ. & Seoul National University
- Monitoring halophyte distribution from multi-period vegetation indices in coastal wetlands. Pending 2025 · Seoul National University
- Determining terrestrialization of abandoned-paddy wetlands using multi-temporal drone LiDAR and stratigraphic volume change. Pending 2023 · Seoul National University
- Deriving optimal wildlife-capture range via machine learning by fusing GPS-tracking data and drone footage. Pending 2023 · Seoul National University

RESEARCH PROJECTS

National R&D and municipal projects (newest first). Budget = total project funding (national R&D; managing agency KEITI). PI = principal investigator.

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|---|-------------------|
| Janghang National Wetland Restoration — ecosystem monitoring | 2025.07 — 2025.11 |
| Korea Environmental Conservation Institute · Researcher · PI: Prof. Youngkeun Song (SNU) | |
| <ul style="list-style-type: none"> • UAV multispectral and LiDAR mapping • Carbon-stock estimation of the site from UAV LiDAR and prediction from the wetland-park vegetation master plan | |
| Carbon storage/sink analysis and inventory for Gyeonggi-do by biotope type | 2024.06 — 2025.05 |
| Gyeonggi Research Institute · Researcher · PI: Prof. Youngkeun Song (SNU) | |
| <ul style="list-style-type: none"> • Analyzed vertical and horizontal forest structure across Gyeonggi Province using ALS • Estimated province-wide forest carbon storage from ALS, airborne hyperspectral, and PlanetScope imagery | |
| Integrated assessment tool for carbon reduction by offset and synergy with ecosystem services | 2023.04 — 2027.12 |
| Korean Ministry of Environment · Researcher · PI: Prof. Youngkeun Song (SNU) · ₩1,579M (total) | |
| <ul style="list-style-type: none"> • Acquired 3D structure of abandoned-paddy wetlands using drone LiDAR • Analyzed carbon storage of abandoned paddies and adjacent vegetation | |
| Gyeonggi-do biotope property-information analysis | 2022.12 — 2023.11 |
| Gyeonggi Research Institute · Researcher · PI: Prof. Youngkeun Song (SNU) | |
| <ul style="list-style-type: none"> • Developed and quantified a method to analyze vegetation layer structure by forest type in Gyeonggi Province | |
| Creation, restoration & management technology of carbon-accumulated abandoned-paddy wetland | 2022.04 — 2026.12 |
| Korean Ministry of Environment · Researcher · PI: Prof. Bonhak Koo (Sangmyung Univ.) · ₩547M (total) | |
| <ul style="list-style-type: none"> • Mapped abandoned-paddy sites and built field drone multispectral / LiDAR datasets | |

Gwacheon-si urban ecological status mapping	2021.09 — 2022.12
<i>Gyeonggi Research Institute · Researcher · PI: Prof. Youngkeun Song (SNU)</i>	
- Classified forest types and tree species of Gwacheon-si forests and urban green spaces - Analyzed vertical and horizontal forest structure of Gwacheon-si	
Real-time web-based positioning surveillance system for introduced exotic species	2021.04 — 2023.12
<i>Korean Ministry of Environment · Project Manager · PI: Prof. Youngkeun Song (SNU) · ₩1,208M (total)</i>	
- Managing researcher — overall scheduling, budget, and research-output management - Attached GPS trackers to four invasive species (red-eared slider, Formosan sika deer, nutria, raccoon) and accumulated movement data - Built drone multispectral, LiDAR, and hyperspectral imagery of invasive-turtle habitats - Produced invasive-species control manuals and developed a real-time positioning-tracking platform	
IT-ET-BT convergence-based distribution and spread models for introduced exotic species	2021.04 — 2023.12
<i>Korean Ministry of Environment · Researcher · PI: Prof. Wanmo Kang (Kookmin Univ.) · ₩650M (total)</i>	
Modeled habitat distribution and spread of invasive turtles from GPS-tracking data using MaxEnt	
Redundancy-based green-infrastructure technology for urban ecosystem challenges	2020.04 — 2022.12
<i>Korean Ministry of Environment · Researcher · PI: Prof. Youngkeun Song (SNU) · ₩927M (total)</i>	
- Installed camera traps to track wild-boar distribution / movement and analyzed home range - Built drone multispectral / LiDAR imagery of wild-boar habitat and performed real-time thermal-UAV tracking	
Monitoring and simulation of climate conditions by spatial type	2020.01 — 2020.08
<i>The Seoul Institute · Researcher · PI: Prof. Youngkeun Song (SNU)</i>	
- Acquired and analyzed field measurements of particulate matter and temperature/humidity by urban spatial type in Seoul - Ran ENVI-met microclimate simulations and designed green-space planting plans for PM reduction, predicting post-implementation outcomes	
Customized habitat-management technique for urban species	2019.04 — 2022.12
<i>Korean Ministry of Environment · Researcher · PI: Prof. Chan Park (Univ. of Seoul) · ₩735M (total)</i>	
Developed a wildlife-detection method using thermal UAV imagery	
Assessment scheme and ecosystem-restoration model by degradation type	2018.07 — 2020.12
<i>Korean Ministry of Environment · Researcher · PI: Prof. Youngkeun Song (SNU) · ₩360M (total)</i>	
Performed land-cover classification using satellite imagery	

AWARDS & HONORS

Awards

Best Presentation Award — KOSERT Autumn Meeting (GEDI vs. airborne LiDAR field agreement by vegetation type and season)	2022
Best Presentation Award — KOSERT Spring Meeting (forest layer-structure indices and their correlation with carbon storage)	2025

Scholarships & Fellowships

Chung Mong-Koo Foundation — OnDream Future Industry Talent Graduate Scholarship (Full)	2022 — Present
BK21 FOUR — Integrated Major in SmartCity Global Convergence (Full)	2020 — Present
Ilju Foundation Undergraduate Scholarship, 23rd Scholar (Full)	2015 — 2017
Challenge Scholarship, Kyungpook National University (Full)	2011 — 2013

PROFESSIONAL SERVICE & MEMBERSHIP

Peer reviewer: International Journal of Applied Earth Observation and Geoinformation (2026 – present)

Member: American Geophysical Union (AGU, 2021–); Ecological Society of America (ESA, 2021–); Korean Society of Environmental Restoration Technology (KOSERT, 2020–); Ecological Society of Korea (2023–)

TEACHING & OUTREACH

University Lecturer (sole instructor) — “Understanding and Application of Spatial Information” (3 cr.; ~30 students/yr), Incheon National University — GIS theory & QGIS labs, remote-sensing theory & analysis	2024, 2025
Online Instructor — 4 GIS/QGIS courses on an online learning platform (1,000+ enrolled students total)	2022 — 2025
Invited Workshops — K-water, Korea Environment Corporation, Hyundai NGV, SeSAC, SNU (20+ sessions)	2020 — 2025

PROFESSIONAL EXPERIENCE

Founder — TREE:ID (startup)	2023.09 — 2025.03
<i>Founded and ran a SaaS startup for urban tree inventory (databasing), diagnosis, and management, Republic of Korea</i>	

Landscape Architect — SUNJIN Engineering & Architecture Co., Ltd. (full-time)

2017.03 — 2018.11

Seoul, Republic of Korea · landscape and urban-planning practice

Design Intern — ATLAS Landscape Architecture, China

2016.09 — 2017.02

China · landscape design practice

TECHNICAL SKILLS & CERTIFICATIONS

Programming & analysis: Python, R, Google Earth Engine

LiDAR / point cloud: ALS · GEDI · MLS/TLS processing, LASTools / PDAL, canopy & vertical-structure metrics

Remote sensing & GIS: QGIS, ArcGIS, satellite & UAV imagery, thermal imaging, Envi-met microclimate

Machine learning: classification / regression (XGBoost, deep learning) for forest & wildlife mapping

Certifications: Engineer Forest, Engineer Landscape Architecture, Drone Pilot License (Class 2)

Languages: Korean (native), English (professional working), Chinese (elementary)